

Data Science and Analytics: Group Exercise

Objective

To assess your organization's data science and analytics capabilities, identify practical opportunities for data science using LLMs, and critically evaluate these opportunities based on data, complexity, and clarity of objectives.

Part 1: Capability Assessment and Opportunity Identification

Instructions:

1. Individual Work (10 minutes)

1.1 Organizational Capability Assessment:

Work individually to rate your organization's current data science and analytics capability on a scale of 1-5, where:

- 1 = Very limited (little or no use of data science/analytics)
- 3 = Developing (some capabilities but not fully integrated or impactful)
- 5 = Advanced (data science is fully embedded and drives decisions).

1.2 Brainstorming Opportunities:

Individually, use a Large Language Model (LLM), such as ChatGPT, to brainstorm practical opportunities for applying data science within your organization. These opportunities should align with your industry and address current business challenges. Focus on identifying opportunities, not success stories.

Be specific in describing these opportunities. For example:

- Weak Example: "Using data science to improve retail operations."
- Strong Example: "Improve inventory planning at the store level by enhancing demand forecasts using customer-level data, coupled with external factors like major events, weather forecasts, promotions, and local competition."

Suggested Prompts to Use with LLMs:

- "What are some examples of using data science in [my industry]?"
 - *Example:* "What are some applications of data science in retail operations?"
- "How can predictive analytics be used to improve [specific area] in my organization?"
 - *Example:* "How can predictive analytics help optimize inventory management?"
- "What are innovative ways to use data in a [size/type of organization] to drive value?"
 - *Example:* "What are data-driven initiatives for a mid-sized healthcare provider?"
- "What kinds of data would I need to implement predictive analytics for [a challenge]?"

- **Example:** "What data is required to predict demand for food supplies in a supermarket chain?"

LLM Brainstorming Tips:

- Be specific about your industry and current business challenges when prompting the LLM.
- Explore both technical (e.g., predictive models, dashboards) and operational opportunities.
- Iterate and refine the outputs to suit your organizational context.

1.3 Idea Selection:

Select one idea that you find most insightful. Use the LLM to refine your thinking further and to frame your opportunity in a way that would be understandable to someone outside your industry.

2. Group Discussion and Selection (20 minutes)

1. Group Capability Average:

Each group member shares their organization's capability score (1-5). Calculate the group's average capability level.

2. Opportunity Sharing and Selection:

- Each group member shares their best opportunity from their individual work.
- As a group, select ONE opportunity that you find the most compelling to present to the main room.

3. Prepare a Short Presentation (2 minutes):

Your presentation should include:

- **The Identified Opportunity**
- **Why It Is Valuable:** Build a business case for why this opportunity matters.
- **What Data or Analytics Capabilities Are Required:** Describe the technical or organizational needs.